

SIMATS ENGINEERING
ASSESSMENT TOOL 2 : Scenario-Based

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

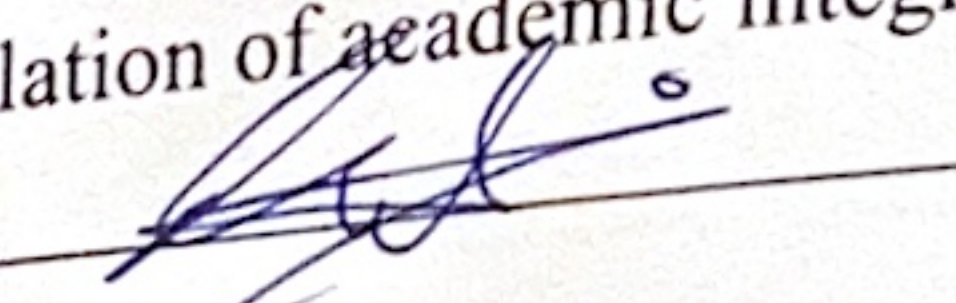
COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I P. Ram Nishu (192511302) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the student: 

Assessment Tasks

Scenario-Based Questions

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.
2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.
3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning.

Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved.
Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.
5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Scenario-Based Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Scenario	20%	Demonstrates complete understanding of the scenario context	Good understanding with minor gaps	Partial understanding	Fails to understand the scenario
Identification of Risks / Issues	20%	Identifies all major issues correctly	Identifies most issues	Limited issue identification	Misses major issues
Application of Concepts	25%	Applies virtualization concepts effectively	Applies concepts with minor mistakes	Partially correct application	Weak application
Decision Justification	20%	Decisions are well justified and technically strong	Reasonable justification	Weak justification	No useful justification
Clarity	15%	Clear and well-organized response	Generally clear	Somewhat unclear	Poorly presented

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Course : Cloud computing and big data analytics

Course code : CSA1521

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1. Media Company Video Streaming platform =>

Introduction

* A media company migrating its video streaming platform to the cloud requires a scalable and highly available architecture to support millions of concurrent user with minimal buffering. Cloud technologies such as caching, load balancing, and auto-scaling help achieve these requirements.

Explanation

* The application should be deployed on multiple cloud servers behind a load balancer to distribute user requests evenly.

* A Content Delivery Network (CDN) caches video content on edge servers located near users, reducing latency & buffering.

* Auto Scaling automatically adds new server instances during peak demand and removes unused instances during low traffic, ensuring optimal performance.

* Cost optimization can be achieved using pay as you go pricing, reserved instances for predictable workloads, and auto-scaling to avoid unnecessary resource usage.

Example

* Netflix uses cloud infrastructure with CDNs, load

balancing and auto-scaling to stream videos smoothly to millions of users worldwide during peak view hrs

Conclusion

* A cloud architecture combining CDN caching, load balancing, auto-scaling and cost optimization techniques ensures high performance, reduced buffering, and efficient resource utilization for video streaming platforms.

2. FinTech startup using Containers

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Introduction

* A FinTech startup deploying applications across multiple cloud providers needs a consistent and efficient deployment strategy. Containerization and orchestration simplify application management while supporting scalability across different cloud environments.

Explanation

* Application should be packed as containers using Docker to ensure consistency.

* Kubernetes can be used for container orchestration to automate deployment, scaling and management.

* Service discovery enables containers to communicate automatically using internal DNS & service registries.

* Kubernetes automatically scales containers based on CPU, memory or user traffic.

* A multi-cloud strategy improves availability and avoids vendor lock-in but introduces challenges such as increased complexity & security management.

Example :

* A digital payment application may run on AWS and Azure simultaneously. Kubernetes manages application deployment across even if one provider experiences an outage.

Conclusion

* Container orchestration with Kubernetes enables consistent deployment, automatic scaling and better resource management while multi-cloud deployment improves reliability and business continuity.

3 E-Learning platform Backup and disaster Recovery

⇒ Introduction

* cloud-based application require backup and disaster recovery mechanisms to protect data against accidental deletion, hardware failures and cyberattacks

Explanation

* cloud-native backup services automatically create scheduled backups of application data.

* Replication stores copies of data in multiple regions to improve availability.

* Versioning maintains previous versions of files, allowing recovery after accidental deletion

* Recovery Time Objective (RTO) is achieved by quickly restoring applications from cloud backups

* Regular disaster recovery testing and automated backup policies ensure business continuity.

Example

* If student records are accidentally deleted administrators can restore the previous version from cloud storage and continue online classes with minimal interruption.

Conclusion

* cloud backup, replication and versioning Provide reliable disaster recovery while achieving low RTO and RPO, ensuring continuous operation of e-learning Platform

4 Government Agency Hybrid cloud ⇒

Introduction

* Government organizations often adopt a hybrid cloud to store sensitive data on-premises while using the public cloud for less sensitive application. This approach balances security, compliance and scalability.

Explanation

* Sensitive government databases remain in the on-premises data center

* Public-facing application such as citizen service portals can run on the public cloud.

* Secure communication between environments is established using VPNs or dedicated private connection

* Encryption protects data both during transmission and while stored.

Example

* A Government department stores confidential citizen records on-premises while hosting its public information website in the cloud. Secure VPN Connection

Conclusion

- * A hybrid cloud provides flexibility, security and regulatory compliance while allowing organizations to utilize cloud resources efficiently for non-sensitive workloads

5 SaaS Provider high availability

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Introduction

- * A SaaS Provider must ensure that its application remains available even if one cloud region experiences a failure. Multi-region deployment provides high availability and fault tolerance.

Explanation

- * Applications should be deployed across multiple cloud regions

- * Global Load balancers distribute user traffic to nearest healthy region

- * Automatic failover redirects traffic to another region if one becomes unavailable

- * Data replication continuously synchronizes databases between regions to ensure data consistency

Example :

- * Microsoft 365 deploys services across multiple regions. If one data center fails, users are automatically redirected to another region with minimal service interruption

Conclusion

• Multi-region, deployment, data replication,
filtering mechanisms and continuous monitoring
Achieve high availability, fault tolerance & encrypted
storage for SaaS applications